Università degli Studi di Napoli

Federico II

Dipartimento di Ingegneria Elettrica e delle Tecnologie dell'Informazione

Statistical Learning and Data Analysis – Module A

ESTIMATION

Prof. Emiliano del Gobbo

emiliano.delgobbo@unina.it

Inference

Statistical inference: *induction* from a sample to a population, under uncertainty to be evaluated in terms of probability.

The **sampling observation** is understood as the realization of a random variable from a probability model on the population which is characterized by model parameters.

Estimation theory: to determine either a point estimation of the model parameters or a confidence interval of values that include the real parameter value.

Testing theory: to verify, empirically, a statement concerning either the model parameters or the probability model class.

In both cases classical statistics considers just one sample but evaluate the uncertainty of the empirical result taking into account all possible "other" results that could be have obtained from all possible "other" samples.

Bayesian statistics add also the prior information about the unknow parameter/model.

© Emiliano del Gobbo - University of Naples Federico II

Estimation

- **Estimator** is a random variable defined as a function of the n-dimensional random sample

$$T_n = t(X_1, \dots, X_n)$$

which elements are independent and identically distributed (*iid*) as the probability model $X \sim f(x, \theta)$ on the population where θ is the unknown parameter to estimate.

- **Estimate** is the value assumed by the estimator for an observed sample:

$$t_n = t(x_1, \dots, x_n)$$

- **Sampling distribution** describes the probability distribution of the estimator for all possible samples (universe of samples).

© Emiliano del Gobbo - University of Naples Federico II

Mean Squared Error

- Error Estimate: $err(T_n) = T_n - \theta$
- Mean Squared Error (MSE):

$$MSE(T_n) = E[T_n - \theta]^2 = var(T_n) + [bias(T_n)]^2$$

where

- $var(T_n) = E[T_n - E(T_n)]^2$
- $bias(T_n) = E(T_n) - \theta$
- Proof:

$$E[T_n - \theta]^2 = E[T_n - E(T_n) + E(T_n) - \theta]^2$$
$$= E[T_n - E(T_n)]^2 + [E(T_n) - \theta]^2 + 2(E(T_n) - \theta)E\big(T_n - E(T_n)\big)$$
$$= var(T_n) + [bias(T_n)]^2$$

- Being $E\big(T_n - E(T_n)\big) = 0$ and $(E(T_n) - \theta)$ a constant.

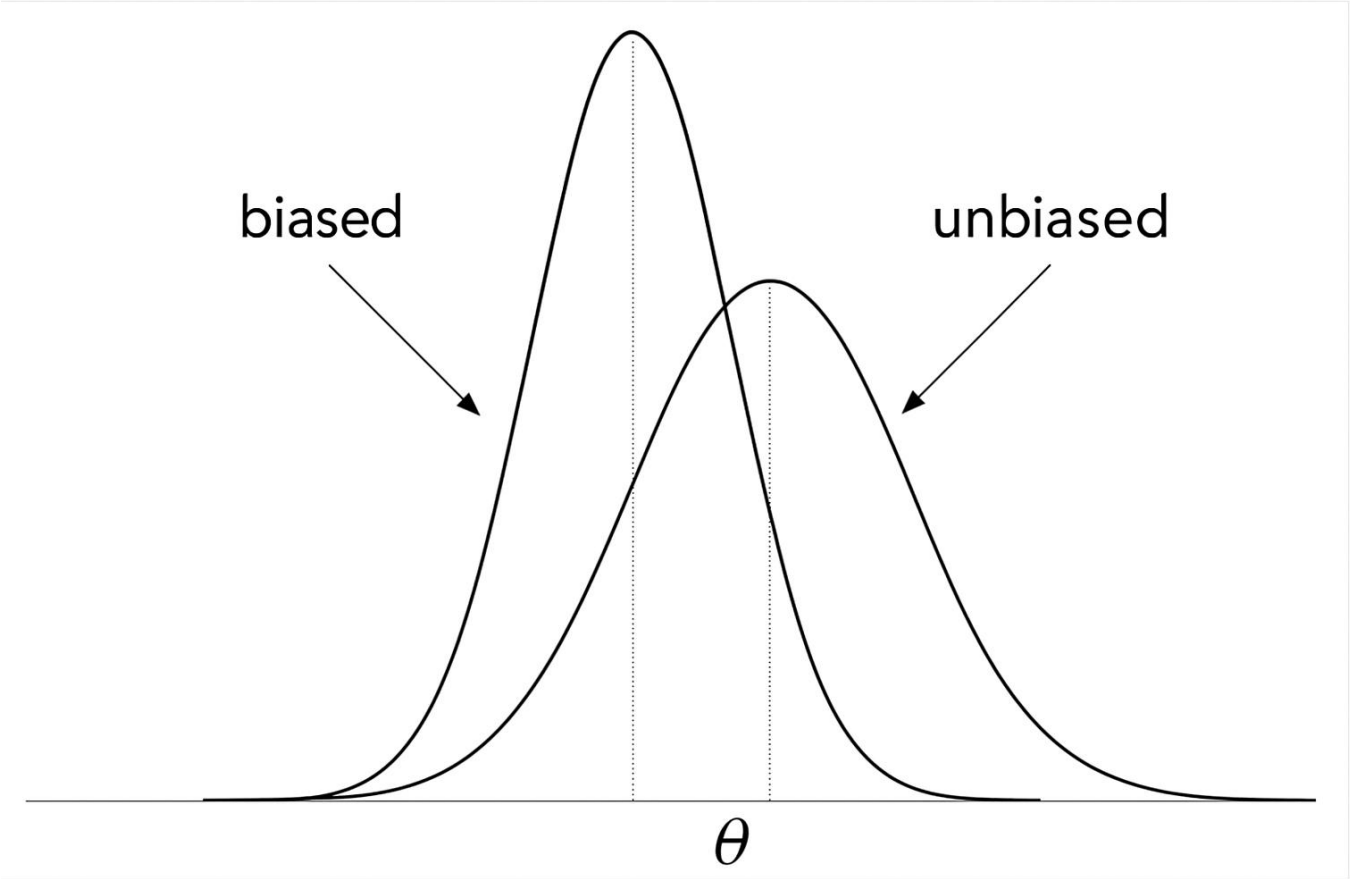
© Emiliano del Gobbo - University of Naples Federico II

Good Estimator Properties for small samples

- T_n is unbiased *iff* $E(T_n) = \theta$
- T_n is relatively more efficient than T_n' *iff* $MSE(T_n) < MSE(T_n')$
- T_n is the most efficient *iff* $MSE(T_n) = \kappa(\theta)$ where $\kappa(\theta)$ is the minimum MSE for an estimator of θ under specific conditions (Frechet, Rao, Cramer, 1943-1946).
  - If it exists an estimator which MSE reaches the minimum of Frechet-Rao-Cramer, this is unique and it is coincident with the Maximum Likelihood Estimator.
- T_n is sufficient when it gives as much information about θ as is possible from the sample at hand.
- Best Linear Unbiased Estimator (BLUE) is an estimator, linear function of the random variables of the sample, unbiased, the most efficient.

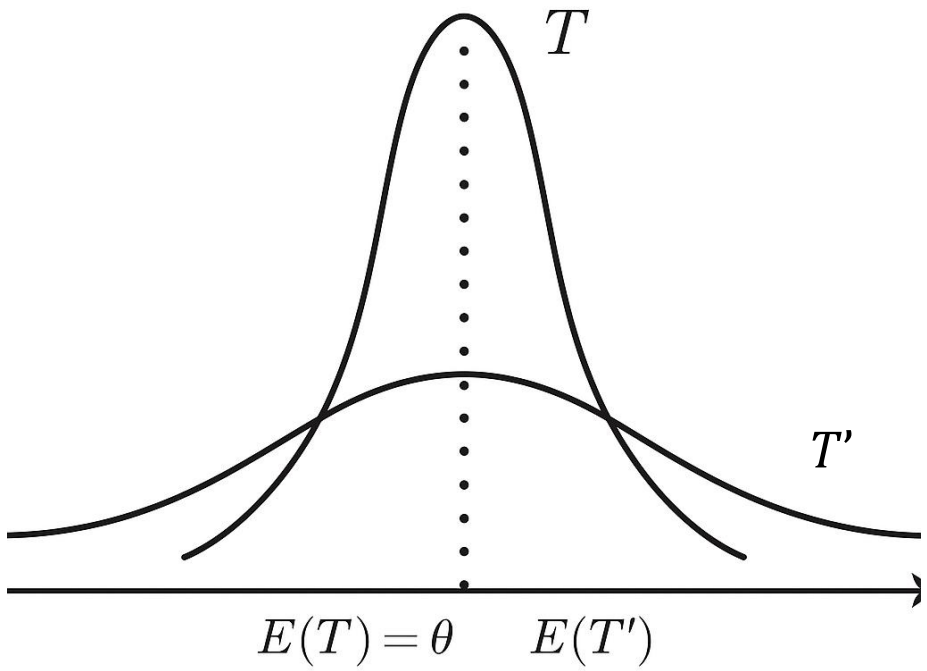
© Emiliano del Gobbo - University of Naples Federico II

Unbiased estimator



© Emiliano del Gobbo - University of Naples Federico II

Efficiency



© Emiliano del Gobbo - University of Naples Federico II

Estimating the Population Mean for a Symmetric Distribution

- Suppose we want to estimate the unknown mean μ of a population with a symmetric distribution.
- Sample Mean:

$$E(\overline{X}) = \mu \quad \text{and} \quad \text{Var}(\overline{X}) = \frac{\sigma^2}{n}$$

- Sample Median:

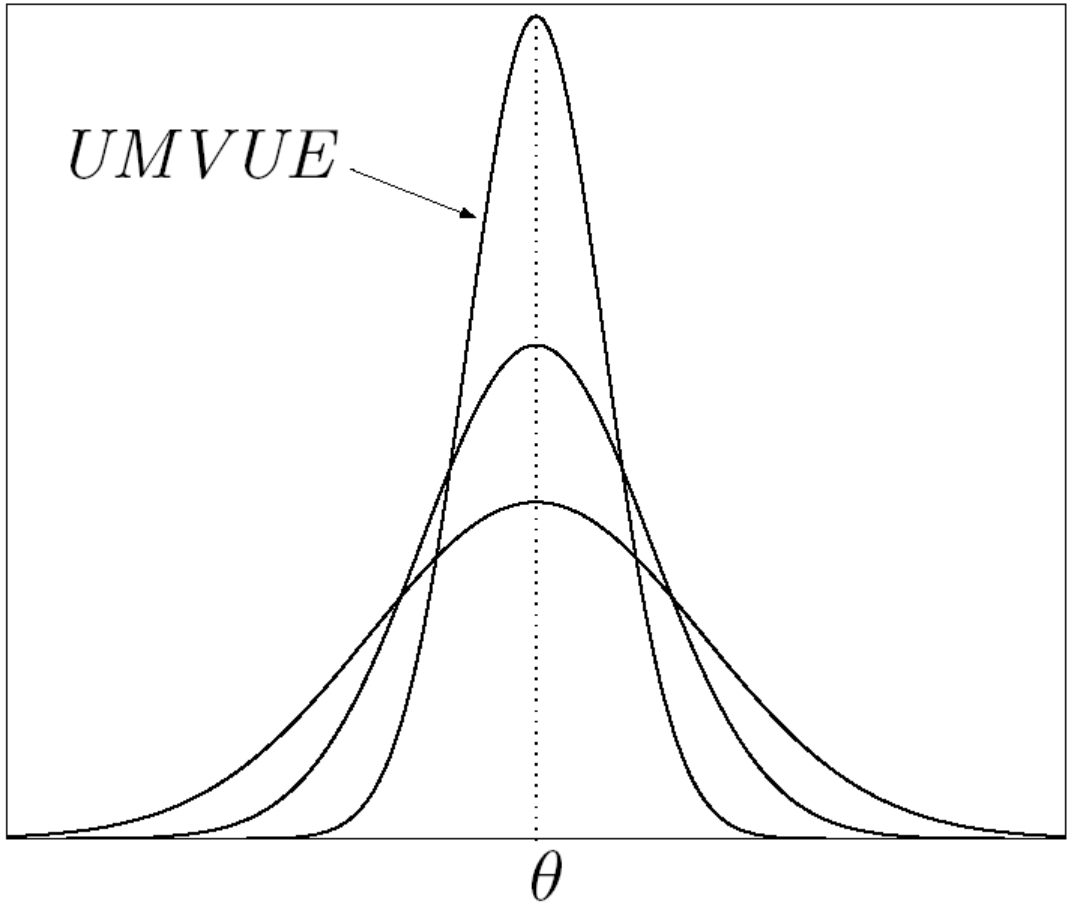
$$E(Me) = \mu \quad \text{and} \quad \text{Var(Me)} = \frac{\sigma^2}{2} \times \frac{\pi^2}{2}$$

- Since:
  $\text{var}(\overline{X}) < \text{var}(\text{ME})$

  the sample mean is chosen, as it is more efficient than the sample median.

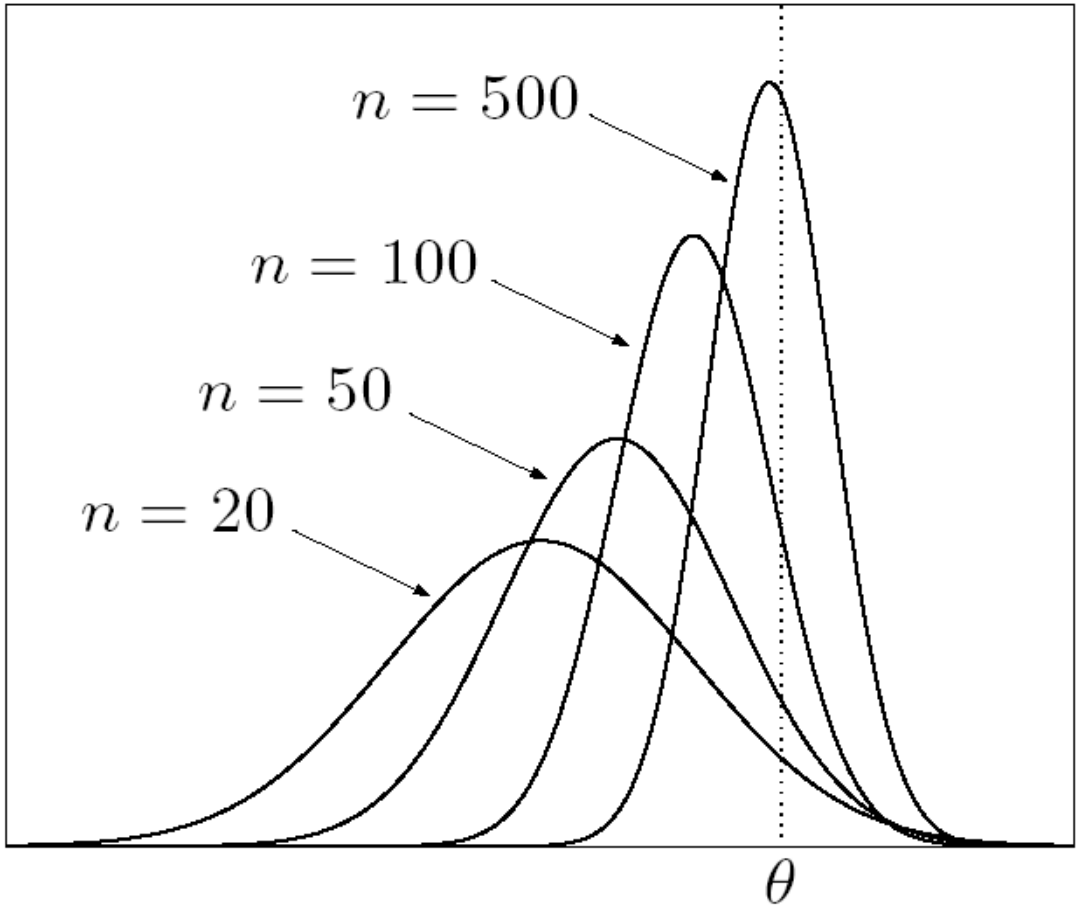
© Emiliano del Gobbo - University of Naples Federico II

UMVUE



© Emiliano del Gobbo - University of Naples Federico II

Consistency



© Emiliano del Gobbo - University of Naples Federico II

Some good estimators

- Sample Mean: $\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$ is Best, Linear, Unbiased, Estimator (BLUE) of the population mean μ of most of the theoretical probability models (uniform, normal, exponential, etc.), with $E(\bar{X}_n) = \mu$ and $var(\bar{X}_n) = \frac{\sigma^2}{n}$.
- Sample Proportion: $P_n = \frac{1}{n}\sum_{i=1}^{n} X_i$ is BLUE of the population mean π of Bernoulli population with $E(P_n) = \pi$ and $var(P_n) = \frac{\pi(1-\pi)}{n}$
- Corrected Sample Variance: $\tilde{S}_n^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X}_n)^2$ is unbiased estimator of the population variance σ^2 since $E\left(\tilde{S}_n^2\right) = \sigma^2$.
- Indeed, $S_n^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X}_n)^2$ is biased since $E(S_n^2) = \frac{n-1}{n}\sigma^2$ so that $\tilde{S}_n^2 = \frac{n}{n-1}S_n^2$ is such that $E\left(\tilde{S}_n^2\right) = E\left(\frac{n}{n-1}S_n^2\right) = \frac{n}{n-1}E(S_n^2) = \frac{n}{n-1}\frac{n-1}{n}\sigma^2 = \sigma^2$.

© Emiliano del Gobbo - University of Naples Federico II

Good Asymptotic Properties (large n sample size)

- T_n is asymptotically unbiased *iff*

$$\lim_{n\to\infty} E(T_n) = \theta$$

- T_n is asymptotically efficient *iff*

$$\lim_{n\to\infty} MSE(T_n) = 0 \Leftrightarrow \lim_{n\to\infty} bias(T_n) = 0 \text{ and } \lim_{n\to\infty} var(T_n) = 0$$

- T_n is consistent *iff*

$$\lim_{n\to\infty} P(|T_n - \theta| < \varepsilon) = 1 \textit{ (weak – convergence in probability)}$$

$$\lim_{n\to\infty} MSE(T_n) = 0 \textit{ (strong – convergence in distribution)}$$

- *If* $\lim_{n\to\infty} MSE(T_n) = 0$ *then* $\lim_{n\to\infty} P(|T_n - \theta| < \varepsilon) = 1$. Vice versa is not necessarily true.
- Proof: Use *Chebychev's inequality*: $P(|T_n - \theta| < k) \geq 1 - \frac{var(T_n)}{k^2}$.

© Emiliano del Gobbo - University of Naples Federico II

Convergence Laws

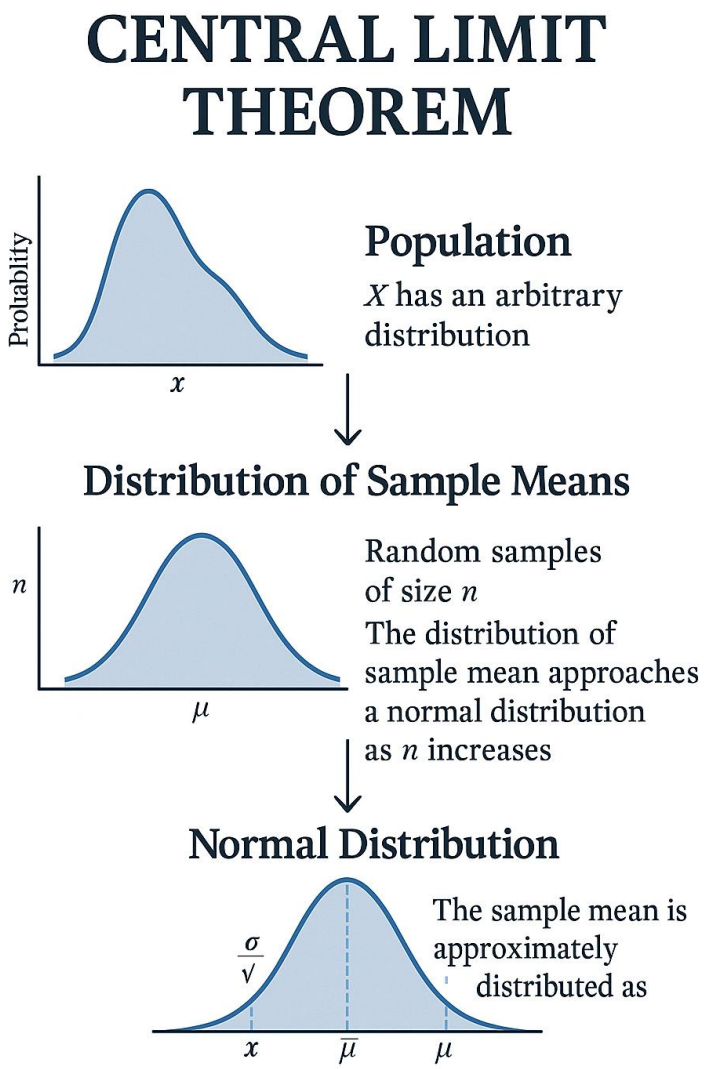
- **Strong Law of large numbers (convergence in distribution):** Performing the same experiment a large number of times, the average of the results should be close to the expected value and will tend to be closer as more trials are performed. The sample average converges *almost surely* to the expected value.
- **Weak Law of large numbers (convergence in probability):** The sample average converges *in probability* to the expected value.

© Emiliano del Gobbo - University of Naples Federico II

Central Limit Theorem

- Let $X_1, X_2, \dots, X_n$ be n random variables, independent and identically distributed (i.i.d.) with mean μ and variance σ^2.
- As n tends to infinite, the sum of all variables tends to a normal distribution with mean μ and variance $n\sigma^2$, as also

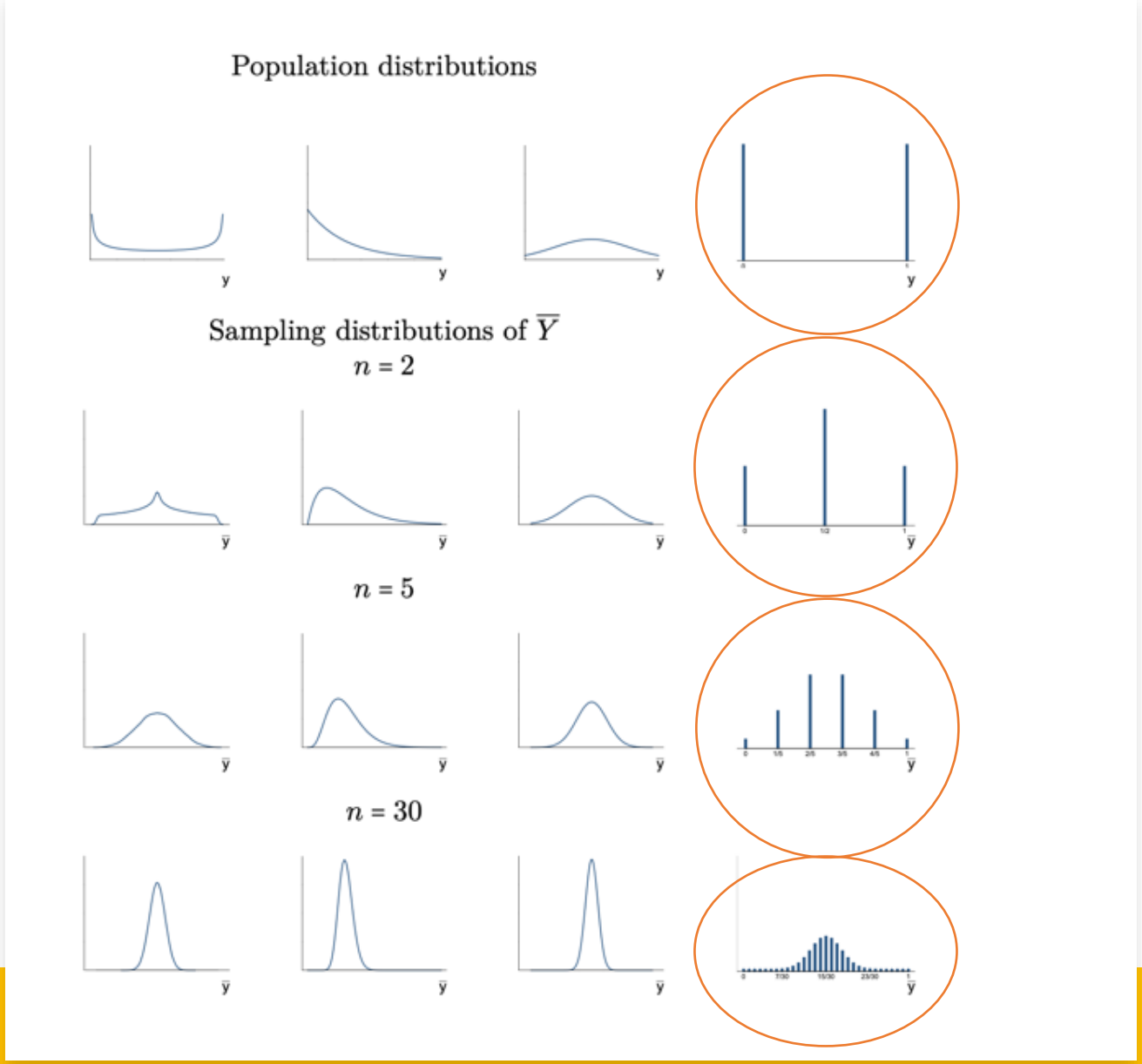
$$\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i \sim N(\mu, \frac{\sigma^2}{n})$$



De Moivre Laplace Theorem

- Let $X_1, X_2, \dots, X_n$ be n Bernoulli random variables, independent and identically distributed (i.i.d.) with mean π and variance $\pi(1-\pi)$.
- As n tends to infinite, the sum of n Bernoulli variables tends to a normal distribution with mean π and variance $n\pi(1-\pi)$, as also

$$\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i \sim N(\pi, \frac{\pi(1-\pi)}{n})$$



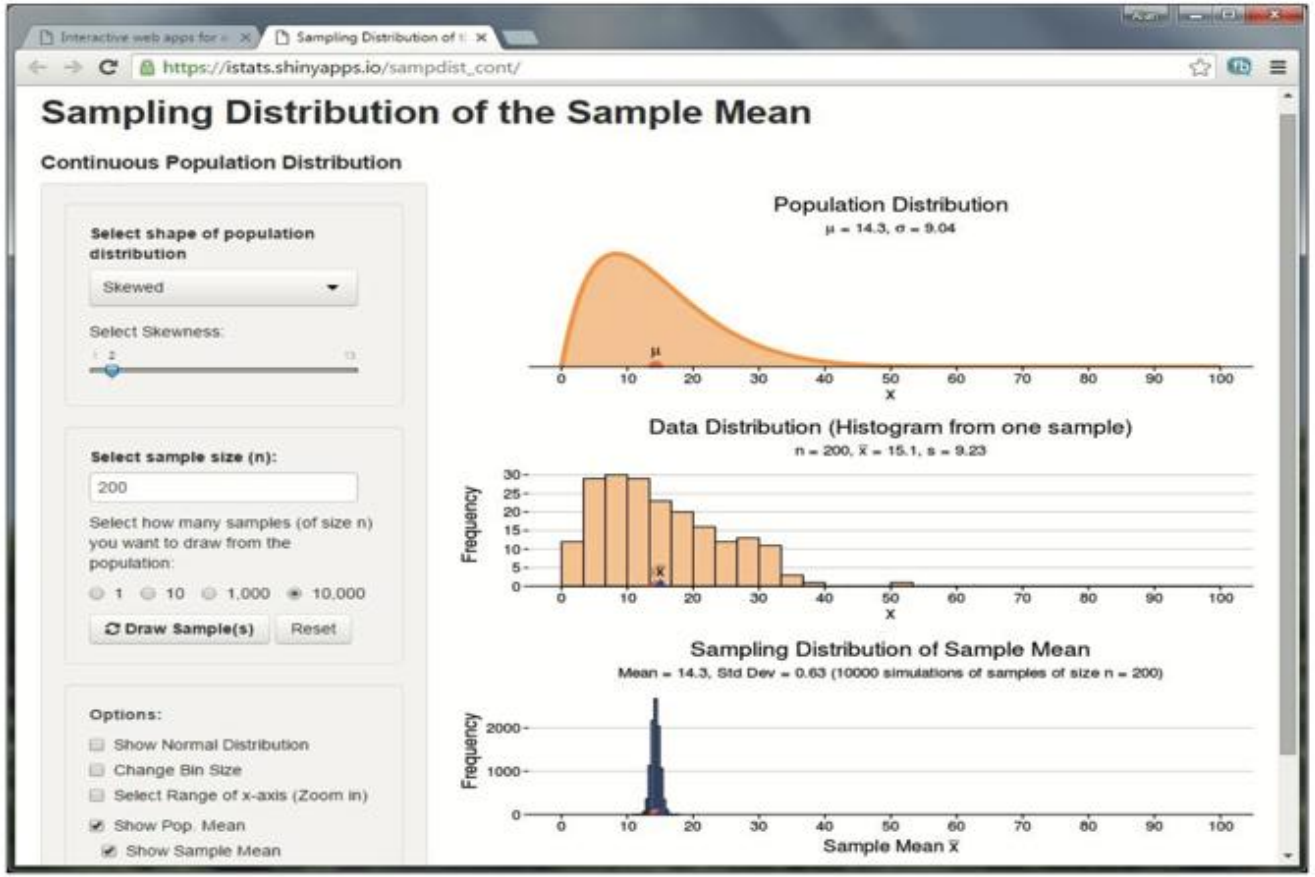


FIGURE 3.5 Screenshot of app at www.artofstat.com for simulating the sampling distribution of $\overline{Y}$ in random sampling from a skewed population distribution: The menu requests 10,000 simple random samples of size n = 200 each (the final sample being shown in the second figure) and plots the 10,000 sample mean values in the bottom figure.

© Emiliano del Gobbo - University of Naples Federico II

Estimation Methods

- Least Squares Estimation
- Maximum Likelihood Estimation
- Bayesian Estimation
- .....

© Emiliano del Gobbo - University of Naples Federico II

Least Squares Estimation

- Let $C_n = (X_1, \dots X_i, \dots, X_n)$ be the n-dimensional random sample which elements X_i are *iid* from an unknown probability model $X \sim f(x, \theta)$ with unknown parameter θ to be estimated.
- Each random variable X_i is as a sum of a deterministic component $g(\theta)$ and an error stochastic component ε_i:

$$X_i = g(\theta) + \varepsilon_i$$

- where $g(\theta)$ is a known function of θ and ε_i are random variable with zero mean, constant variance and they are uncorrelated.
- **Least Squares (LS) Estimate** is obtained by

$$\min_{\theta \in \Theta} S(\theta) = \sum_{i=1}^{n} \left(x_i - g(\theta)\right)^2$$

- For $g(\theta) = \theta$, the problem is to estimate the real value of a quantity upon n measurements of that quantity.
- The **LS Estimate** of the population mean μ is the sample mean $\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$ for the sample realization. **LS Estimator** of the population mean μ is a random variable $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$. LSE are BLUE.

© Emiliano del Gobbo - University of Naples Federico II

The Likelihood Function

- Let $X \sim f(x;\theta)$ be the known probability model on the population where θ is the unknown parameter to estimate with $\theta \in \Theta$.
- Let $C_n = (X_1, \dots X_i, \dots, X_n)$ be the n-dimensional random sample which elements X_i are *iid* as the population probability model $X \sim f(x;\theta)$.
- Define the **Likelihood Function (LF)** of the random sample as follows:

$$l(X_1, \dots, X_n|\theta) = \prod_{i=1}^{n} f(X_i, \theta)$$

- which provides the joint probability of the random sample drawn from a known population probability model characterized by the parameter θ.
- The LF is function of both the random sample and the unknown parameter.

© Emiliano del Gobbo - University of Naples Federico II

Maximum Likelihood Estimator

- For an observed sample $c_n = (x_1, \dots x_i, \dots, x_n)$, the LF is function of only the unknown parameter.
- The Maximum Likelihood (ML) estimate is found by maximizing the LF over all possible parameters:

$$\max_{\theta \in \Theta} l(\theta) = l(x_1, \dots x_i, \dots, x_n | \theta) = \prod_{i=1}^{n} f(x_i, \theta) = l(\hat{\theta})$$

- $l(\theta)$quantifies the likelihood to obtain that observed sample for any value of θ. $l(\hat{\theta})$ says that $\hat{\theta}$ is the value of the parameter that maximizes this likelihood.
- The ML estimator $\hat{\theta}_{ML}$ can be defined as a function of the n-dimensional random sample considering the closed formula derivation of the maximum likelihood estimation.
- The LF is usually a globally convex and monotone function so that maximizing the LF is equivalent to maximizing the Log-Likelihood Function $L(\theta)$.

© Emiliano del Gobbo - University of Naples Federico II

ML Estimation of the probability of success

- $X \sim B(1, \pi)$ be the population Bernoulli probability model with unknown parameter π to be estimated.
- The probability model distribution is given by $P(X = x) = \pi^x (1 - \pi)^{1-x}$, where $x = 0,1$.
- The Maximum Likelihood (ML) estimate can be found as follows:

$$\max_{\pi \in \Pi} L(\pi) = l(x_1, \dots x_i, \dots, x_n | \pi) = \prod_{i=1}^{n} \pi^{x_i} (1 - \pi)^{x_i} = \pi^{\sum_{i=1}^{n} x_i} (1 - \pi)^{n - \sum_{i=1}^{n} x_i}$$

- Which is equivalent to maximizing:

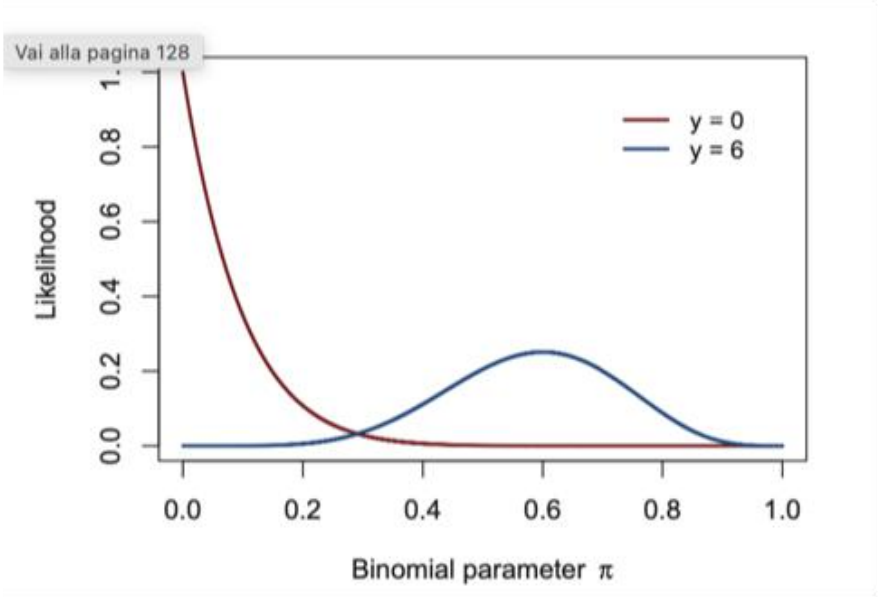
$$L(x_1, \dots x_i, \dots, x_n | \pi) = \sum_{i=1}^{n} x_i \log \pi + (n - \sum_{i=1}^{n} x_i) \log(1 - \pi)$$

- The ML Estimate is equal to $p_{ML} = \frac{1}{n} \sum_{i=1}^{n} x_i$.
- In general, for any other observed realization of the random sample, the ML Estimator is the Sample Proportion $P_n = \frac{1}{n} \sum_{i=1}^{n} X_i$.

© Emiliano del Gobbo - University of Naples Federico II

Example

- $X \sim B(1, \pi)$ be the population Bernoulli probability model with unknown parameter π to be estimated.
- The probability model distribution is given by $P(X = x) = \pi^x (1 - \pi)^{1-x}$, where $x = 0,1$.
- Let $c_{10} = (1, 0, 0, 0, 1, 1, 1, 1, 0, 1)$ be the observed sample.
- The Likelihood Function provides the joint probability to obtain exactly the observed sample given any parameter π:
- $l(1, 0, 0, 0, 1, 1, 1, 1, 0, 1|\pi) = \prod_{i=1}^{10} \pi^{x_i}(1 - \pi)^{x_i} = \pi^{\sum_{i=1}^{10} x_i}(1 - \pi)^{10-\sum_{i=1}^{10} x_i} = \pi^6(1 - \pi)^4$
- The ML Estimate is equal to $p_{ML} = \frac{6}{10} = 0.6$.



Agresti (2022) Foundations of Statistics for Data Scientists

Properties of ML Estimator

- ML estimators have desirable asymptotic (i.e., large n) properties:

✓ML estimators are asymptotically unbiased: as n increases, any bias they have diminishes to 0.

✓ML estimators are consistent: as n increases, the estimator converges toward the parameter value.

✓ML estimators are asymptotically efficient: for large n, other estimators do not have smaller standard errors and tend to fall closer to the parameter.

✓ML estimators have asymptotic normal sampling distributions.

✓ML estimators are invariant to linear transformation.

✓

© Emiliano del Gobbo - University of Naples Federico II

Fisher's Information

- Let $C_n = (X_1, \dots X_i, \dots, X_n)$ be the n-dimensional random sample which elements X_i are *iid* as the population probability model $X \sim f(x;\theta)$.
- Fisher's Information is defined as $I(\theta) = nE\left(\frac{\partial log f(X;\theta)}{\partial \theta}\right)^2$

where the Expectation is taken with respect to X for a fixed θ. The information increases as the sample size n increases.

- Fisher showed that the approximate normal distribution of the ML estimator $\hat{\theta}_{ML}$ is $\hat{\theta}_{ML} \sim N\left(\theta, \frac{1}{I(\theta)}\right)$.
- The variance decreases as n and thus the information increase. Any estimator that has variance that simplifies to $1/I(\theta)$, for large n is asymptotically efficient.
- For the Bernoulli population, $I(\pi) = \frac{n}{\pi(1-\pi)}$ so that $\hat{\pi}_{ML} \sim N\left(\pi, \frac{\pi(1-\pi)}{n}\right)$. The Sample Proportion P_n is MLE, asymptotically normally distributed.
- For the Normal population, $I(\mu) = \frac{n}{\sigma^2}$ so that $\hat{\mu}_{ML} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$. The Sample Mean $\bar{X}_n$ is MLE, asymptotically normally distributed.

© Emiliano del Gobbo - University of Naples Federico II

Precision of ML Estimator

- Under certain conditions, an alternative expression for Fisher's Information make use of the log-likelihood function $L(\theta)$:

$$I(\theta) = -nE\left(\frac{\partial^2 log f(X;\theta)}{\partial\theta^2}\right) = -E\left(\frac{\partial^2 L(\theta)}{\partial\theta^2}\right)$$

- Since the second derivative of a function describes curvature, the corresponding alternative expression $1/I(\theta)$ for the large-sample variance of $\hat{\theta}$ indicates that $var(\hat{\theta})$ is smaller when we can expect the log-likelihood function to be more highly curved.
- In summary, the log- likelihood function is important not only to identify the $\hat{\theta}$ value that maximizes it but also to use its curvature to determine the **precision** of $\hat{\theta}$ as an estimator of θ.

© Emiliano del Gobbo - University of Naples Federico II

Sufficient Estimator

- The estimator $T_n(\mathbf{X}) = T_n = t(X_1, \dots, X_n)$ is defined as a sufficient statistic for θ if the conditional distribution of $\mathbf{X}|T(\mathbf{X})$ does not depend on θ. There is no more information about θ from the data $\mathbf{x} = (x_1, \dots, x_n)$ once that $T_n(\mathbf{X})$ is known.
- For the most commonly-used distributions, an estimator $T_n(\mathbf{X}) = T_n = t(X_1, \dots, X_n)$ exists such that the likelihood function factors for an observed sample can be expressed as
$$l(\theta) = g(T_n(\mathbf{x}); \theta) h\big(T_n(\mathbf{x})\big)$$
- The function involving θ depends on the observed data $\mathbf{x} = (x_1, \dots, x_n)$ only through $T_n(\mathbf{x})$ whereas the separate function involving $\mathbf{x}$ does not involve θ.
- Finding θ to maximize $l(\theta)$ is equivalent to finding θ to maximize $g(T_n(\mathbf{x}); \theta)$.
- So, the ML estimate of θ depends on the data only through $T_n(\mathbf{x})$, not each individual observation.
- The statistic $T_n(\mathbf{X})$ is called the sufficient statistic for θ. Once we know its value, the data provide no further information about θ.
- The Rao-Blackwell Theorem says that if $T_n{}'(\mathbf{X})$ is not a sufficient statistic for θ a more efficient estimator exists that is obtained by conditioning on the sufficient statistic.

© Emiliano del Gobbo - University of Naples Federico II

Bayes Estimator

- Let $X \sim f(x;\theta)$ be the known probability model on the population where θ is the unknown parameter to estimate with $\theta \in \Theta$.
- Let θ be a random variable with prior probability distribution $g(\theta)$.
- Let $C_n = (X_1, \dots X_i, \dots, X_n)$ be the n-dimensional random sample which elements X_i are *iid* as the population probability model $X \sim f(x;\theta)$.
- The Bayes estimate is found by maximizing the posterior probability over all possible parameters:

$$\max_{\theta \in \Theta} P(\theta | x_1, \dots x_i, \dots, x_n) \propto g(\theta) l(x_1, \dots x_i, \dots, x_n | \theta)$$

- Large sample properties of MLE extends to Bayesian estimators. As n increases, the posterior distribution of θ is mainly affected by the likelihood function.

© Emiliano del Gobbo - University of Naples Federico II

Inference on the population mean μ

- The **random variable Sample Mean** $\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$ is the estimator of the population mean μ.
- Let σ^2 be the variance of the population.
- The Sample Mean is normally distributed

$$\bar{X}_n \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

  - ✓ if the population is assumed to be normally distributed *(typical assumption for small sample size)*
  - ✓ for the central limit theorem for large sample size.

- Its standardized form has a standardized normal distribution:

$$Z_n = \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \sim N(0,1)$$

- If the variance σ^2 is unknown, it can be estimated by the unbiased sample variance $\tilde{S}_n^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X}_n)^2$ yielding to the t-Student distribution:

$$T_n = \frac{\bar{X}_n - \mu}{\tilde{s}_n/\sqrt{n}} \sim t_{n-1}$$

- For large samples ($n > 30$) the t-Student converges to the $N(0,1)$.

© Emiliano del Gobbo - University of Naples Federico II

Standard Error

- The standard error of the sampling distribution of Sample Mean is given by $\sigma_{\bar{X}} = \sqrt{var(\bar{X}_n)} = \frac{\sigma}{\sqrt{n}}$.
- It tells about the variability of all possible estimates from the real value μ when changing the observed sample.
- For sampling without replacement from a finite population, it reduces to $\sigma_{\bar{X}} = \sqrt{var(\bar{X}_n)} = \frac{\sigma}{\sqrt{n}}\sqrt{\frac{N-n}{N-1}}$

© Emiliano del Gobbo - University of Naples Federico II

Confidence Interval of the population mean μ with σ aknown

- **Confidence interval** can be built up using the following equation and the probability tables of the standardized normal distribution:

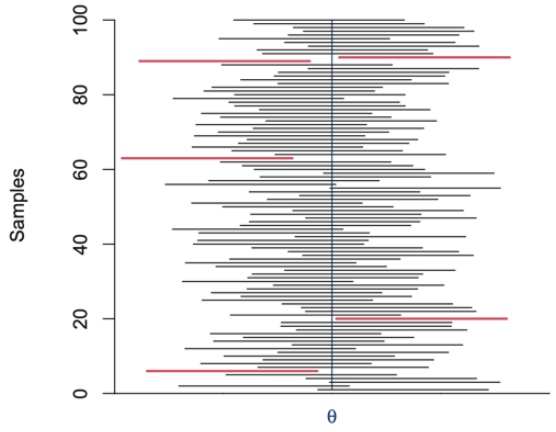
$$P\left(-z_{\alpha/2} \leq Z \leq +z_{\alpha/2}\right) = 1 - \alpha$$

- where $z_{\alpha/2}$ is such that $P\left(|Z| > \left|z_{\alpha/2}\right|\right) = \alpha/2$.
- The confidence interval for the unknown parameter μ can be derived as
  - ***Before sampling***: it is a probability

$$P\left(\bar{X}_n - z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X}_n + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right) = 1 - \alpha$$

  - ***After sampling***: it is an interval estimate

$$\left[\bar{x}_n \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right] \text{ with confidence level } (1-\alpha)\%$$

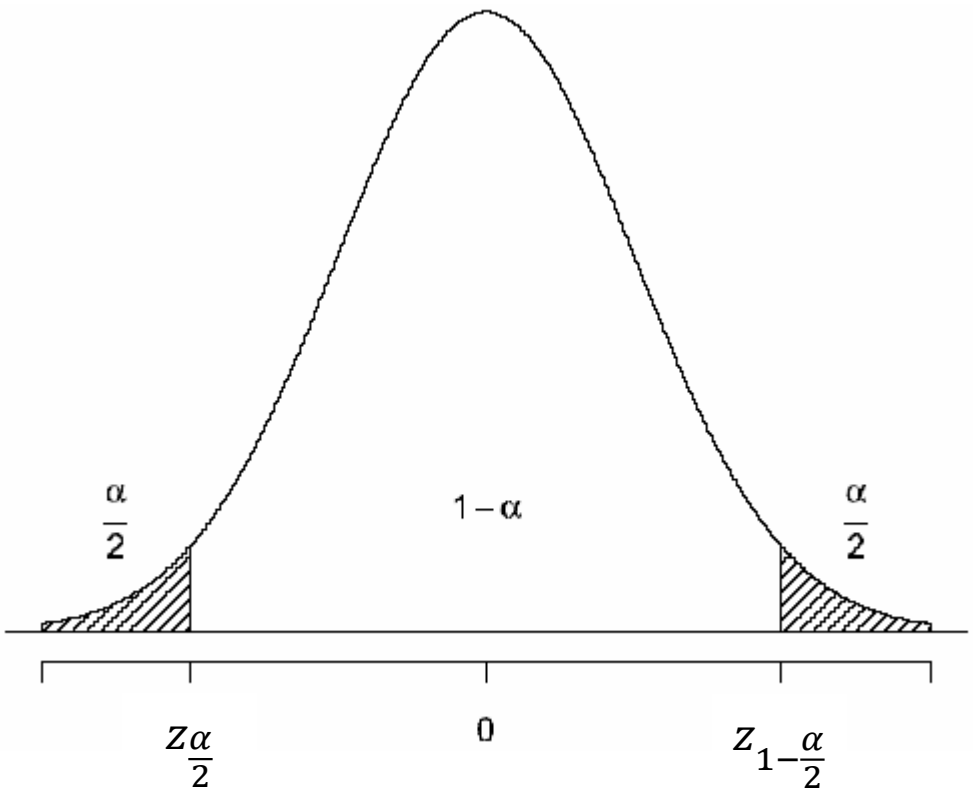


Universe of Confidence Intervals

- If the variance σ^2 is unknown, it can be estimated by the unbiased sample variance and then use the t-Student distribution. Note that for large samples ($n > 30$) the t-Student converges to the $N(0,1)$.

© Emiliano del Gobbo - University of Naples Federico II

Confidence Interval: sampling mean



© Emiliano del Gobbo - University of Naples Federico II

Confidence Interval of the population mean μ with σ unknown

- **Confidence interval** can be built up using the following equation and the probability tables of the Student T distribution:

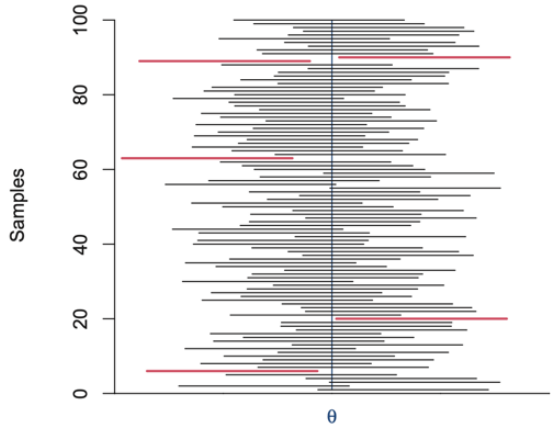
$$P\left(-\mathrm{t}_{\mathrm{n}-1,\alpha/2} \leq \mathrm{T} \leq +\mathrm{t}_{\mathrm{n}-1,\alpha/2}\right) = 1-\alpha$$

- where $\mathrm{t}_{\alpha/2}$ is such that $P\left(|\mathrm{T}| > \left|\mathrm{t}_{\mathrm{n}-1,\alpha/2}\right|\right) = \alpha/2$.
- The confidence interval for the unknown parameter μ can be derived as
  - ***Before sampling***: it is a probability

$$P\left(\bar{X}_n - \mathrm{t}_{\mathrm{n}-1,\alpha/2}\frac{\mathrm{s}}{\sqrt{n}} \leq \mu \leq \bar{X}_n + \mathrm{t}_{\mathrm{n}-1,\alpha/2}\frac{\mathrm{s}}{\sqrt{n}}\right) = 1-\alpha$$

  - ***After sampling***: it is an interval estimate

$$\left[\bar{x}_n \pm \mathrm{t}_{\alpha/2}\frac{\mathrm{s}}{\sqrt{n}}\right] \text{ with confidence level } (1-\alpha)\%$$



Universe of Confidence Intervals

- If the variance σ^2 is unknown, it can be estimated by the unbiased sample variance and then use the t-Student distribution. Note that for large samples ($n > 30$) the t-Student converges to the $N(0,1)$.

© Emiliano del Gobbo - University of Naples Federico II

Key elements in confidence intervals

- The interval estimate of the population mean μ:

$$\left[\bar{x}_n \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right] \text{ with confidence level } (1-\alpha)\%$$

- **Key elements**:
  - ✓ Variance of the population: σ^2
  - ✓ Sample size: n
  - ✓ Interval size or error estimate: $\varepsilon = z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$
  - ✓ Confidence level: $1-\alpha$
- One can fix three over four elements and derive the fourth from the confidence interval equation.

© Emiliano del Gobbo - University of Naples Federico II

Confidence Interval of the population variance σ (small samples)

Unbiased sample variance

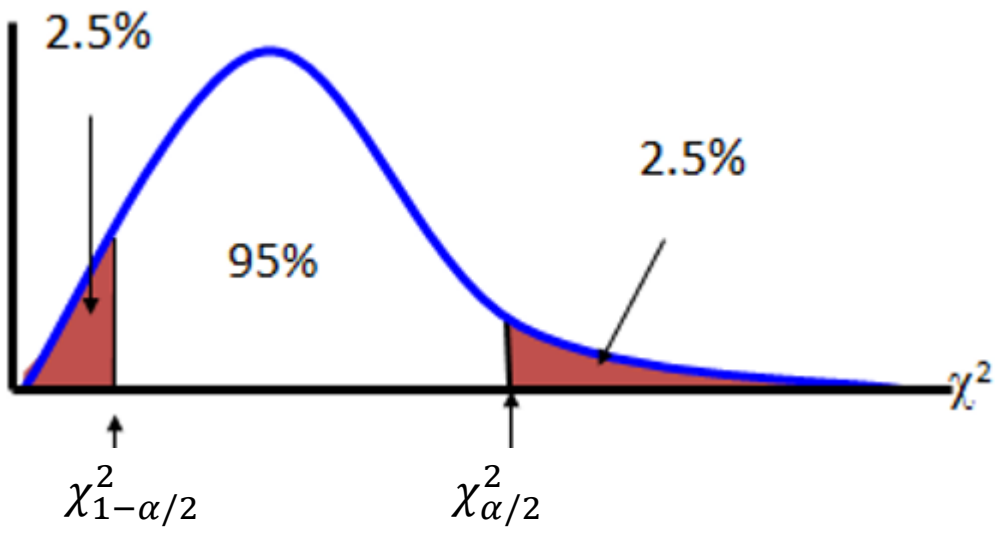
$$S_c^2 = S_n^2 = \frac{\sum_{i=1}^{N}(X_i - \bar{X})^2}{n-1}$$

The estimate of the sample variance for small samples follows a chi-squared χ^2 distribution with $n-1$ gradi di libertà. degrees of freedom. Therefore, for a given significance level α, we have:

$$\chi^2 = \frac{(n-1)S_n^2}{\sigma^2} \qquad P\left(\chi^2_{1-\frac{\alpha}{2}} < \frac{(n-1)S_n^2}{\sigma^2} < \chi^2_{\frac{\alpha}{2}}\right) = 1-\alpha \qquad \chi^2_{1-\frac{\alpha}{2}} < \frac{(n-1)S_n^2}{\sigma^2} < \chi^2_{\frac{\alpha}{2}}$$

And through appropriate steps:

$$\frac{(n-1)\cdot s_n^2}{\chi^2_{\frac{\alpha}{2}}} < \sigma^2 < \frac{(n-1)\cdot s_n^2}{\chi^2_{1-\frac{\alpha}{2}}}$$



© Emiliano del Gobbo - University of Naples Federico II

Confidence Interval of the population variance σ (big samples)

For large samples, the estimate of the sample variance follows a normal distribution, according to the Central Limit Theorem.

$$Z = \frac{S - \sigma}{\frac{\sigma}{\sqrt{2n}}} \sim N(0,1)$$

$$-z_{\frac{\alpha}{2}} < \frac{S - \sigma}{\frac{\sigma}{\sqrt{2n}}} < z_{\frac{\alpha}{2}}$$

Therefore, for a given significance level α, and through appropriate steps:

$$\frac{s}{1 + \frac{z_{\frac{\alpha}{2}}}{\sqrt{2n}}} < \sigma < \frac{s}{1 - \frac{z_{\frac{\alpha}{2}}}{\sqrt{2n}}}$$

© Emiliano del Gobbo - University of Naples Federico II